

Linear Algebra II (2026 Fall)

Lecture 5.

Studying Contents :

Chapter 5. Diagonalization

- Section 5.2 : Diagonalizability

Section 5.2 Diagonalizability

Example 4. : Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by

$$T \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 4a_1 & & + a_3 \\ 2a_1 + 3a_2 + 2a_3 \\ a_1 & & + 4a_3 \end{pmatrix}.$$

In this example, we want to find the eigenspaces corresponding to each eigenvalue of T .

(1) Let $\beta = \{e_1, e_2, e_3\}$ be the standard basis for $\mathbb{R}^3$. Then

$$[T]_{\beta} = \begin{pmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{pmatrix},$$

and hence the characteristic polynomial of T is

$$\det([T]_{\beta} - tI) = \det \begin{pmatrix} 4-t & 0 & 1 \\ 2 & 3-t & 2 \\ 1 & 0 & 4-t \end{pmatrix} = -(t-5)(t-3)^2.$$

So, the eigenvalues of T are $\lambda_1 = 5$ and $\lambda_2 = 3$ with multiplicities 1 and 2, respectively.

Example 4. (Continued) :

(2) Since

$$E_{\lambda_1} = N(T - \lambda_1 I) = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 : \begin{pmatrix} -1 & 0 & 1 \\ 2 & -2 & 2 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\},$$

it follows that E_{λ_1} is the solution space of the system of linear equations

$$\begin{aligned} -x_1 &+ x_3 = 0 \\ 2x_1 - 2x_2 + 2x_3 &= 0 \\ x_1 &- x_3 = 0. \end{aligned}$$

One can check easily that

$$E_{\lambda_1} = \left\{ c \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \mid c \in F \right\} \text{ and hence } \dim E_{\lambda_1} = 1 \text{ (= the multiplicity of } \lambda_1 = 5).$$

(3) Since $E_{\lambda_2} = N([T]_{\beta} - \lambda_2 I)$ is the solution space of the system of linear equations

$$\begin{aligned} x_1 + x_3 &= 0 \\ 2x_1 + 2x_3 &= 0 \\ x_1 + x_3 &= 0 \end{aligned}$$

it follows that

$$E_{\lambda_2} = \left\{ c \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + d \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \mid c, d \in F \right\} \text{ and hence } \dim E_{\lambda_2} = 2 \text{ (= the multiplicity of } \lambda_2 = 3).$$

Example 4. (Continued) :

(4) The following set of eigenvectors of T is a basis for $\mathbb{R}^3$.

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Consequently, T is diagonalizable.

Remark : In Example 3,

(i) the characteristic polynomial of T splits, but

(ii) the dimension of an eigenspace is strictly less than the multiplicity.

In this case, T fails to be diagonalizable.

In Example 4,

(i) the characteristic polynomial of T splits, and

(ii) the dimension of each eigenspace is equal to the multiplicity of the corresponding eigenvalue.

In this case, T fails to be diagonalizable.

This is indeed true, as we now show.

Theorem 5.8. : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

Also let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of T .

For each $1 \leq i \leq k$, let $S_i \subset E_{\lambda_i}$ be a finite linearly independent subset.

Then $S = S_1 \cup S_2 \cup \dots \cup S_k$ is a linearly independent subset of V .

To prove Theorem 5.8, we need the following lemma.

Lemma : Let $T : V \rightarrow V$ and $\lambda_1, \dots, \lambda_k$ be as in Theorem 5.8.

For each $i = 1, 2, \dots, k$, let $v_i \in E_{\lambda_i}$.

If $v_1 + v_2 + \dots + v_k = O_V$, then $v_1 = v_2 = \dots = O_V$

Lemma : Let $T : V \rightarrow V$ and $\lambda_1, \dots, \lambda_k$ be as in Theorem 5.8.

For each $i = 1, 2, \dots, k$, let $v_i \in E_{\lambda_i}$.

If $v_1 + v_2 + \dots + v_k = O_V$, then $v_1 = v_2 = \dots = v_k = O_V$

Proof : Suppose not.

We may assume that $v_1, \dots, v_m$ are nonzero and $v_{m+1} = \dots = v_k = O_V$ for some $m \geq 1$.

Thus $v_1 + v_2 + \dots + v_m = O_V$ and $v_1, \dots, v_m$ are linearly independent by Theorem 5.5.

This is a contradiction, and hence it must be true that $v_1 = v_2 = \dots = v_k = O_V$. ■

Theorem 5.5 : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

Let $\lambda_1, \dots, \lambda_k \in \mathbb{F}$ be distinct eigenvalues of T and

let $v_1, \dots, v_k \in V$ be eigenvectors of T such that $Tv_j = \lambda_j v_j$ ($1 \leq j \leq k$).

Then $\{v_1, \dots, v_k\}$ is linearly independent.

Theorem 5.8. : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

Also let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of T .

For each $1 \leq i \leq k$, let $S_i \subset E_{\lambda_i}$ be a finite linearly independent subset.

Then $S = S_1 \cup S_2 \cup \dots \cup S_k$ is a linearly independent subset of V .

Proof : For each i , let $S_i = \{v_{i1}, v_{i2}, \dots, v_{in_i}\}$. Thus $S = \{v_{ij} \mid 1 \leq i \leq k, 1 \leq j \leq n_i\}$.

If $\sum_{1 \leq i \leq k, 1 \leq j \leq n_i} a_{ij}v_{ij} = O_V$, then

$$\sum_{1 \leq i \leq k} \left(\sum_{1 \leq j \leq n_i} a_{ij}v_{ij} \right) = O_V.$$

Note that $\sum_{1 \leq j \leq n_i} a_{ij}v_{ij}$ is a member of E_{λ_i} for each $1 \leq i \leq k$.

Therefore, by the previous Lemma, it holds that

$$\sum_{1 \leq j \leq n_i} a_{ij}v_{ij} = O_V \text{ for each } 1 \leq i \leq k.$$

Then $a_{ij} = 0$ for all $1 \leq j \leq n_i$ since S_i is linearly independent.

We conclude that $S = S_1 \cup S_2 \cup \dots \cup S_k$ is linearly independent. ■

Theorem 5.9. : Let $T : V \rightarrow V$ be a linear operator on an n -dimensional vector space V whose characteristic polynomial of T splits and let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T . Then

- (1) $T : V \rightarrow V$ is diagonalizable if and only if the multiplicity of λ_i is equal to $\dim E_{\lambda_i}$ for all i .
- (2) Suppose T is diagonalizable and β_i is an ordered basis for E_{λ_i} for each i . Then

$$\beta := \beta_1 \cup \dots \cup \beta_k$$

is an ordered basis for V consisting of eigenvectors of T .

Proof : (1) For each i , let m_i denote the multiplicity of λ_i and $d_i = \dim E_{\lambda_i}$.

($\Rightarrow$) Suppose that T is diagonalizable. Let β be a basis for V consisting of eigenvectors of T .

Let $\beta_i = \beta \cap E_{\lambda_i}$, the set of vectors in β that are eigenvectors corresponding to λ_i .

Then $|\beta_i| \leq d_i$ since β_i is a linearly independent subset of the d_i -dimensional space E_{λ_i} .

Also $d_i \leq m_i$ by Theorem 5.7. Thus we have

$$n = |\beta| = \sum_{i=1}^k |\beta_i| \leq \sum_{i=1}^k d_i \leq \sum_{i=1}^k m_i = n.$$

It follows that $|\beta_i| = d_i = m_i$ for all $1 \leq i \leq k$.

Theorem 5.9. : Let $T : V \rightarrow V$ be a linear operator on an n -dimensional vector space V whose characteristic polynomial of T splits and let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T . Then

(1) $T : V \rightarrow V$ is diagonalizable if and only if the multiplicity of λ_i is equal to $\dim E_{\lambda_i}$ for all i .

(2) Suppose T is diagonalizable and β_i is an ordered basis for E_{λ_i} for each i . Then

$$\beta := \beta_1 \cup \dots \cup \beta_k$$

is an ordered basis for V consisting of eigenvectors of T .

Proof : (1) For each i , let m_i denote the multiplicity of λ_i and $d_i = \dim E_{\lambda_i}$.

($\Leftarrow$) Suppose that $d_i = m_i$ for all $1 \leq i \leq k$.

Let β_i be an ordered basis for E_{λ_i} , and let $\beta = \beta_1 \cup \dots \cup \beta_k$.

Then $|\beta| = |\beta_1| + \dots + |\beta_k| = m_1 + \dots + m_k = n$.

Also, Theorem 5.8 says that β is linearly independent.

Thus β is a basis for V , and hence $T : V \rightarrow V$ is diagonalizable.

(2) This is already proved in the proof of (1). ■

© Test for diagonalization

Let $T : V \rightarrow V$ be a linear operator on an n -dimensional vector space V . Then

T is diagonalizable **if and only if**

- (i) The characteristic polynomial of T splits; and
- (ii) For each eigenvalue λ of T , the multiplicity of λ equals $n - \text{rank}(T - \lambda I)$.

Remark 1. : If T is diagonalizable and $\beta_1, \dots, \beta_k$ are ordered bases for the eigenspaces of T , then

the union $\beta = \beta_1 \cup \dots \cup \beta_k$ is an ordered basis for V consisting of eigenvectors of T , and hence $[T]_\beta$ is a diagonal matrix.

Step 1. : Choose a “convenient” basis α for V and let $B := [T]_\alpha$.

Step 2. : Test if the characteristic polynomial $f(t) = \det(B - tI_n)$ of T splits.

If $f(t)$ does **not split**, then T is **not diagonalizable**.

Step 3. : Suppose that $f(t)$ splits. Then check if

the multiplicity of each “repeated” eigenvalue of B equals $n - \text{rank}(B - \lambda I)$

By Theorem 5.7, condition (ii) is automatically satisfied for eigenvalues of multiplicity 1.

If so, then B , and hence T , is diagonalizable.

Step 4. : Suppose that T is diagonalizable.

Then we first find a basis for each eigenspace of B .

The union of these bases is a basis γ for F^n consisting of eigenvectors of B .

Each vector in γ is the coordinate vector relative to α of an eigenvector of T .

$$[\]_{\alpha} : V \rightarrow F^n, \gamma = \{u_1, \dots, u_n\}$$

$$\Rightarrow \text{There exist } v_1, \dots, v_n \in V \text{ such that } [v_i]_{\alpha} = u_i.$$

$$\Rightarrow [T(v_i)]_{\alpha} = [T]_{\alpha}[v_i]_{\alpha} = Bu_i = \lambda u_i = [\lambda v_i]_{\alpha} \text{ and hence } T(v_i) = \lambda v_i.$$

The set consisting of these n eigenvectors of T is

the desired basis β for V consisting of eigenvectors of T .

Example 5. : We test the following matrix for diagonalizability.

$$A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix} \in M_{3 \times 3}(R)$$

(1) The characteristic polynomial of A is $\det(A - tI) = -(t - 4)(t - 3)^2$, which splits.

Also A has eigenvalues $\lambda_1 = 4$ and $\lambda_2 = 3$ with multiplicities 1 and 2, respectively.

(2) ($\lambda_1 = 4$) **Theorem 5.7** says that $1 \leq \dim E_{\lambda_1} \leq 1$, and so $\dim E_{\lambda_1} = 1$.

This means that $\text{rank}(A - \lambda_1 I) = 3 - \dim E_{\lambda_1} = 3 - 1 = 2$, and hence

$$3 - \text{rank}(A - \lambda_1 I) = 3 - 2 = 1 \text{ is equal to the multiplicity of } \lambda_1.$$

Thus condition (ii) is satisfied for λ_1 .

(3) ($\lambda_2 = 3$) Because the matrix $A - \lambda_2 I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ has rank 2, we see that

$$3 - \text{rank}(A - \lambda_2 I) = 3 - 2 = 1, \text{ which is not the multiplicity of } \lambda_2.$$

Therefore A **is not diagonalizable**.

Example 6. : Let $T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ be the linear operator defined by

$$T(f(x)) = f(1) + f'(0)x + (f'(0) + f''(0))x^2.$$

(1) For $\alpha = \{1, x, x^2\}$, we have

$$T(1) = 1, \quad T(x) = 1 + x + x^2 \quad \text{and} \quad T(x^2) = 1 + 2x^2.$$

Thus it follows that

$$B := [T]_{\alpha} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}.$$

Then the characteristic polynomial of B (and hence of T) is $-(t-1)^2(t-2)$, which splits.

Also B has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$ with multiplicities 2 and 1, respectively.

(2) Condition (ii) is satisfied for λ_2 because it has multiplicity 1.

For $\lambda_1 = 1$, we have

$$3 - \text{rank}(B - \lambda_1 I) = 3 - \text{rank} \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix} = 3 - 1 = 2,$$

which is equal to the multiplicity of λ_1 .

Therefore T is diagonalizable.

Example 6. (Continued) : (3) The eigenspace corresponding to $\lambda_1 = 1$ is

$$E_{\lambda_1} = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 : \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\},$$

which is the solution space for the linear equation $x_2 + x_3 = 0$.

Therefore we have $E_{\lambda_1} = \left\{ a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \mid a, b \in \mathbb{R} \right\}.$

Also $\gamma_1 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \right\}$ is a basis for E_{λ_1} .

(4) The eigenspace corresponding to $\lambda_2 = 2$ is

$$E_{\lambda_2} = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 : \begin{pmatrix} -1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\},$$

which is the solution space for the system of linear equations

$$\begin{cases} -x_1 + x_2 + x_3 = 0 \\ x_2 = 0 \end{cases}.$$

Therefore we have $E_{\lambda_2} = \left\{ c \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \mid c \in \mathbb{R} \right\}$. Also $\gamma_2 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for E_{λ_2} .

Example 6. (Continued) : (5) Let

$$\gamma := \gamma_1 \cup \gamma_2 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Then γ is an ordered basis for $\mathbb{R}^3$ consisting of eigenvectors of B .

(6) Consider the coordinate map $[\]_\alpha : P_2(\mathbb{R}) \rightarrow \mathbb{R}^3$, $f(x) = a + bx + cx^2 \mapsto (a, b, c)$.

Then one can observe that the vectors in γ are

the coordinate vectors relative to α of the vectors in the set $\beta = \{1, -x + x^2, 1 + x^2\}$.

In consequence, β is an ordered basis for $P_2(\mathbb{R})$ consisting of eigenvectors of T .

Also, it follows that

$$[T]_\beta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Example 7. : Find A^n for all $n \geq 1$ where

$$A = \begin{pmatrix} 0 & -2 \\ 1 & 3 \end{pmatrix}.$$

(1) $\det(A - tI) = (t-1)(t-2)$, and hence A has two distinct eigenvalues, $\lambda_1 = 1$ and $\lambda_2 = 2$.

(2) $A - I = \begin{pmatrix} -1 & -2 \\ 1 & 2 \end{pmatrix}$ and hence $u = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ is an eigenvector of A corresponding to $\lambda_1 = 1$.

$A - 2I = \begin{pmatrix} -2 & -2 \\ 1 & 1 \end{pmatrix}$ and hence $v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an eigenvector of A corresponding to $\lambda_2 = 2$.

(3) Since $Au = u$ and $Av = 2v$, we have $A[u \ v] = [u \ v] \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$.

Thus it holds that

$$A = \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & -2 \end{pmatrix}.$$

(4) For every $n \geq 1$, it follows that

$$A^n = \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2^n \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} 2 & 2^n \\ -1 & -2^n \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} 2-2^n & 2-2^{n+1} \\ -1+2^n & -1+2^{n+1} \end{pmatrix}.$$